

ECEN-629: Homework 2

Due on September 30, 2025 at 11:59 pm

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Problem 1

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is concave, and $a, b \in \mathbf{dom} f$ with $a < b$.

(a) Prove the following inequality:

$$\frac{f(x) - f(a)}{x - a} \geq \frac{f(b) - f(a)}{b - a} \geq \frac{f(b) - f(x)}{b - x}.$$

for any $x \in (a, b)$.

(b) For $f = -x^2 + 2x + 5$, $a = 0$, and $b = 3$, draw a sketch to illustrate the inequalities above.

Solution

Problem 2

Show that the maximum of a convex function f over the polyhedron $\mathcal{P} = \mathbf{conv}\{v_1, v_2, \dots, v_k\}$ is achieved at one of the vertices, that is

$$\sup_{x \in \mathcal{P}} f(x) = \max_{i=1,2,\dots,k} f(v_i).$$

In particular, let $f(x) = x_1^2 + 2x_1x_2 + 3x_2^2$, and let $v_1 = (1, 1)^T$, $v_2 = (1, -1)^T$, $v_3 = (1, 1)^T$, and $v_4 = (-1, -1)^T$. Compute its maximum over the polyhedron.

Solution

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Problem 3

Suppose

$$f(x) = h(g_1(x), g_2(x)),$$

where $g_1(x) = \|x\|_2^2$, $g_2(x) = \|x\|_1$, and $h(u, v) = \log(e^u + e^v)$.

- (a) Show $f(x)$ is convex by applying the correct composition rule.
- (b) Give another h that keeps f convex but is non-logarithmic, and explain why.

Solution

Problem 4

Let $f_1, f_2, \dots, f_m$ be convex function on $\mathbb{R}^n$. Define

$$f(x) = \inf \{ f_1(x_1) + f_2(x_2) + \dots + f_m(x_m) \mid x_i \in \mathbb{R}^n, x_1 + x_2 + \dots + x_m = x \}.$$

Show $f(x)$ is convex on $\mathbb{R}^n$.

Solution

Problem 5

The r -separation function $\Delta_r(x)$ of the vector $x \in \mathbb{R}^n$, where $r \leq n$, is defined as the difference between the sum of the r largest components of the vector x and the sum of the r smallest components.

- (a) Is the function $\Delta_r(x)$ convex? Prove your answer.
- (b) Let $r = 1$ and $n = 2$. Sketch the function $\Delta_1(x)$ in the positive quadrant.

Solution

Problem 6

Suppose $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex, and $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $d \in \mathbb{R}$. Show that the function that $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, where

$$g(x) = (c^T x + d) f\left(\frac{Ax + b}{c^T x + d}\right), \quad \text{dom } g = \{x \mid c^T x + d > 0\}.$$

Provide a clear argument which rules you are applying, and the exact manner of function composition.

Solution

Problem 7

A vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called monotone if $\langle f(x) - f(y), x - y \rangle \geq 0$. Show that the gradient of a convex function is monotone.

Solution

Problem 8

The proximal operator of a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ is defined by

$$\text{prox}_{\lambda f}(v) = \arg \min_x \left(f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right).$$

- (a) Is the proximal operator a convex function?
- (b) Derive the proximal operator for $f(x) = \|x\|_1$.
- (c) Solve explicitly for $v = (3, -2, 0.5)^T$ and $\lambda = 1$ when $f(x) = \|x\|_1$.

Solution

Problem 9

Given a norm $\|\cdot\|$ on $\mathbb{R}^n$, the dual norm is defined as

$$\|y\|_* = \sup_{\|x\| \leq 1} y^T x.$$

- (a) Show that $\|\cdot\|_*$ is indeed a norm.
- (b) Compute the dual norm of $\|x\|_1$ and $\|x\|_\infty$.
- (c) For $x = (1, 2)^T$, compute $\|x\|_2^*$.

Solution